TraceChain

Supply-Chain Provenance & Custody Protocol on BOT Chain

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PROTOCOL OVERVIEW

TraceChain is a supply-chain provenance protocol deployed on BOT Chain that records every custody transfer and status update for a product from origin to delivery. Admin-approved participants (manufacturers, logistics providers, distributors, retailers) register products and log on-chain updates at each custody handoff. Any third party — consumer, regulator, auditor — can trace the full journey of any product by ID without a wallet. The result is an immutable, public, tamper-proof audit trail from source to shelf.

A technical overview of the protocol, its design, core functions, security properties, and roadmap.

1. Problem Statement

• Counterfeit goods — $500B+ in counterfeit products enter supply chains annually; consumers have no reliable way to verify authenticity.

• Opaque provenance — food, pharma, and luxury goods change hands multiple times with no persistent audit trail.

• Siloed tracking systems — each participant uses a different ERP; data is not shared or cross-verifiable.

• Recall inefficiency — when a defect is discovered, tracing affected batches takes days or weeks with paper records.

• No public access — consumers and regulators cannot independently verify supply chain claims.

2. Protocol Design

2.1 Participant Roles

`Role`
Permissions

Admin

Register / deactivate participants; transfer admin role

Participant (active)

Register products; log updates; transfer custody; mark delivery

Current Holder

Only the current holder of a product can add updates or transfer custody

Public

Track any product by ID — no wallet, no registration

2.2 Provenance Trail

Each product has an ordered Step[] trail — an append-only array of location, status, note, actor address, and timestamp. Steps can only be added by the current holder. Custody transfers change the holder and append a step atomically. Once delivered, no further updates are accepted.

3. Contract Architecture

3.1 Data Model

Participant: name, role (string), active flag.

Product: name, batch, origin, creator address, current holder address, createdAt

timestamp, delivered flag.

`Step: by (address), location, status, note, time.`

3.2 Write Functions

`Function`
`Role`
Description

`addParticipant(addr, name, role)`
Admin

Registers a supply chain participant

`removeParticipant(addr)`
Admin

Deactivates a participant

`transferAdmin(addr)`
Admin

Transfers admin role

`registerProduct(name, batch, origin)`
Participant

Creates a product — caller becomes first holder

`addUpdate(id, location, status, note)`
Current Holder

Logs a status update without custody change

`transferCustody(id, to, location, status, note)`
Current Holder

Transfers product to another participant

markDelivered(id, location, note)

Current Holder

Marks product as finally delivered

3.3 Read Functions

`Function`
`Returns`
`getProduct(id)`
name, batch, origin, creator, holder, createdAt, delivered

`trailLength(id)`
Number of steps in the product's trail

`getStep(id, index)`
by, location, status, note, time for a single step

`productCount()`
Total number of registered products

`Function`
`Returns`
`participantAt(i)`
Address, name, role, active for a participant

4. Security Properties

4.1 Holder-Only Updates

onlyHolder(id) requires products[id].holder == msg.sender && !products[id].delivered. A participant who

has transferred custody cannot add further updates to that product.

4.2 Delivery Finality

Once products[id].delivered = true, the onlyHolder modifier blocks all further updates. The trail is

permanently frozen.

4.3 Participant Gating

`transferCustody() requires participants[to].active — custody cannot be transferred to an unregistered or` deactivated address.

4.4 Append-Only Trail

The Step[] trail is append-only. There is no function to edit, delete, or reorder steps. The entire history is permanently visible on-chain.

5. Events

Event	Parameters
ParticipantAdded	who, name, role
ParticipantRemoved	who
ProductRegistered	id, creator
StepAdded	id, by, status
CustodyTransferred	id, from, to

6. Use Cases

Industry	Application
Food & Agriculture	Farm-to-shelf traceability; cold-chain monitoring
Pharmaceuticals	Batch tracking; recall efficiency; anti-counterfeiting
Luxury Goods	Provenance certificates; anti-counterfeit authentication
Electronics	Component sourcing; recycling chain tracking
Logistics	Cross-border shipment visibility; customs audit trails

7. Deployment

Parameter	Value
Chain	BOT Chain (Chain ID 968)
RPC	https://rpc.botchain.ai
Explorer	https://scan.botchain.ai
Public tracking	

No wallet required to read product trail

8. Roadmap

Milestone	Description
v1.1	IoT sensor data attestation (signed location pings appended as steps)
v1.2	QR-code deep-link generator for consumer-facing product pages
v1.3	Batch operations — register and transfer multiple products in one tx
v2.0	Cross-chain provenance bridge for multi-chain supply networks